Curriculum Vitae

MINBEOM KIM

Ph. D., Civil and Environmental Engineering, The Pennsylvania State University (Penn State, USA)

Email: mmbb1805@gmail.com, Phone: (+82) 10-8865-0694, Google Scholar

EDUCATION

Ph.D., Department of Environment and Energy Engineering, GIST	Mar 2020 – Aug 2025
M.S., Department of Environment and Energy Engineering, GIST	Mar 2018 – Feb 2020
B.S., Department of Materials Science and Engineering, GIST	Mar 2014 – Feb 2018
Busan Il Science High School (one-year early graduation)	Mar 2012 – Feb 2014

EXPERIENCE

Post Doc., Department of Civil and Environmental Engineering, Penn State	Nov 2026 – Present
Post Doc., Department of Environment and Energy Engineering, GIST	Aug 2025 – Oct 2026
Research Assistant, Department of Environment and Energy Engineering, GIST	Mar 2018 – Aug 2025
Teaching Assistant, Environmental Nanotechnology (*Course Title*), GIST	Mar 2023 – Aug 2023
Teaching Assistant, Environmental and Energy Engineering (*Course Title*), GIST	Mar 2020 – Aug 2020
Research Project Manager, Environmental Nanotechnology Lab, GIST	Mar 2018 – Oct 2026

- "Highly efficient nano-composite membrane-based ONE-WATER circulation technology development for AIR-AQUA NEXUS in future cities (Sub: Development of high-performance nanofiber composite membranes for white plume and ultrafine-dust seed substances)", National Research Foundation of Korea, Ministry of Science and ICT, Republic of Korea, *USD 2,880,000* (2020-2025).
- "Development of thin film composite electrospun nanofiber membrane for water vapor separation", GIST Research Institute, Ministry of Science and ICT, Republic of Korea, *USD 7,200* (2023-2023).
- "Development of electrospun nanofiber composite conductive films for EMI shielding", Korea Technology and Information Promotion Agency for SMEs, Ministry of SMEs and Startups, Republic of Korea, *USD 360,000* (2018-2021).
- "Development and commercialization of safe electrospun nanofiber mask filter by using non-toxic organic solvent", National Research Foundation of Korea, Ministry of Science and ICT, Republic of Korea, *USD 36,000* (2020-2021).
- “Development of multi-functional electrospun nanofiber composite film for Electromagnetic interference shielding”, GIST Research Institute, Ministry of Science and ICT, Republic of Korea, *USD 7,200* (2020-2020).
- "Development of functional nanofiber filter for microplastic treatment in water", National Research Foundation of Korea, Ministry of Science and ICT, Republic of Korea, *USD 91,300* (2019-2020).

Participating Researcher, Environmental Nanotechnology Lab, GIST	Mar 2018 – Oct 2026

- "Development of artificial intelligence-based precision medicine and healthcare technologies", GIST Research Institute, Ministry of Science and ICT, Republic of Korea, *USD 108,800* (2025-Present).
- "Photothermal energy absorbing nanofiber-based super-evaporator development for next-generation Zero Liquid Discharge (ZLD) system", National Research Foundation of Korea, Ministry of Science and ICT, Republic of Korea, *USD 432,000* (2024-Present).
- "Environmental convergence technology for virtuous resource circulation society", GIST Research Institute, Ministry of Science and ICT, Republic of Korea, *USD 7,200* (2025-2025).
- "Environmental and energy solutions for a sustainable future", GIST Research Institute, Ministry of Science and ICT, Republic of Korea, *USD 7,200* (2024-2024).
- "Amphiphilic nanofiber filter with tailored wettability for oil emulsion separation", National Research Foundation of Korea, Ministry of Science and ICT, Republic of Korea, *USD 432,000* (2020-2023).
- "Development of flame-retardant electrospun nanofiber-based thermal insulation material for energy efficiency and fire response plan ", GIST Research Institute, Ministry of Science and ICT, Republic of Korea, *USD 7,200* (2022-2022).
- "Organic solvent resistant thin-film composite hollow fiber membranes based on polymers of intrinsic microporosity for fine chemical industries", GIST Research Institute, Ministry of Science and ICT, Republic of Korea, *USD 7,200* (2021-2022).
- "Development of antifouling nanocomposite hollow fiber membrane for energy harvesting using wastewater", GIST Research Institute, Ministry of Science and ICT, Republic of Korea, *USD 7,200* (2019-2020).

Lab Manager, Environmental Nanotechnology Lab, GIST — Mar 2021 – Mar 2024

RESEARCH EXPERIENCE

Hybrid Nanofibrous Membranes for Environmental Pollution Control

- Necklace-Like MOF/Nanofiber Membranes for Integrated Particulate and Gaseous Pollutant Removal
- In-Situ Crystallization of MOFs on Functionalized Nanofibers for Antibiotic Removal in Aquatic Systems
- Carbonaceous Layer-by-Layer Nanofiber Composites for Selective Water Vapor Transport and Separation
- Electrospun Hybrid Nanofibrous Platforms for Advanced Electromagnetic Interference (EMI) Shielding

Structure–Property Engineering of Hybrid Nanomaterials

- Surface-functionalization and Tailored Surface Chemistry of Nanomaterials for Selective Pollutant Capture
- Hierarchically Porous Architectures and Interfacial Compatibility in MOF-Polymer Composite
- Long-term Stability, Scalability, and Energy-Efficient Membrane Process in Environmental Remediation
- MOF/COF Composite Membranes for Electrochemical Ion Separation and PFAS-specific Mitigation

- Ph.D. dissertation thesis: Hybrid Nanofibrous Multifunctional Composite Membranes for Environmental Pollution Control: A Synergistic Design of MOFs, Carbon, and Polymer Composites. (Advisor: Prof. Heechul Choi, GIST)
- M.S. dissertation thesis: Fabrication of carbon nanotube buckypaper/polymer nanofiber composite hybrid film for high-performance electromagnetic interference (EMI) shielding. (Advisor: Prof. Heechul Choi, GIST)

PUBLICATIONS

Journals

Minbeom Kim, Hyeonseo Kim, Jaehyun Byun, Eunmok Yang, and Heechul Choi. "Gravity-driven removal of tetracycline from water using a hierarchically porous adsorptive nanofibrous membrane system functionalized with metal–organic framework." *Water Research* (2026): 124939.

Minbeom Kim, Eunmok Yang, Yejin Liang, Soyoung Kim, Jaehyun Byun, Hyeonseo Kim, and Heechul Choi. "Rational design of a necklace-like ZIF-67/poly (vinylidene fluoride) electrospun nanofiber hybrid membrane for simultaneous removal of $PM_{0.3}$ and SO_2." *ACS Applied Materials & Interfaces* 16.12 (2024): 15348-15361.

Minbeom Kim, Soyoung Kim, Yu Chang Seong, Kwang-Hoon Yang, and Heechul Choi, "Multiwalled carbon nanotube buckypaper/polyacrylonitrile nanofiber composite membranes for electromagnetic interference shielding." *ACS Applied Nano Materials* 4.1 (2021): 729-738.

Jaehyun Byun, **Minbeom Kim (Co-first author)**, Eunmok Yang, Hyeonseo Kim, and Heechul Choi. "Electrospun nanofiber-supported thin-film composite membrane by introducing graphene oxide interlayer for water vapor/N_2 separation." *Journal of Environmental Chemical Engineering* (2024): 114409.

Jaehyun Byun, Hyeonseo Kim, Andrey Kim, **Minbeom Kim* (Co-corresponding author)**, and Heechul Choi*. " Integrated electrospinning and electrospray fabrication of nanofiber-supported polyamide membranes with GO and GO/GOQD interlayers for efficient water-selective air dehumidification." *Under review in Journal of Industrial Engineering and Chemistry* (2026)

Minbeom Kim and Heechul Choi. "Hierarchically porous MOF composite electrospun nanofibrous membrane for efficient PFAS removal from water." *In preparation* (2026)

Jangho Lee, Kwonjung Lee, Eunmok Yang, **Minbeom Kim**, Jiwon Seo, and In S. Kim. " Sequence-Defined Peptoid–Silica Sorbents for Dialysate Regeneration." *In preparation* (2026)

Hyeonseo Kim, **Minbeom Kim**, Jaehyun Byun, Eunmok Yang, In S. Kim, and Heechul Choi. "Dual nanofiber membrane with asymmetric wettability for high-performance solar evaporator in zero liquid discharge system." *Chemical Engineering Journal* (2026): 172738.

Eunmok Yang, **Minbeom Kim**, Yejin Liang, Jaehyun Byun, Hyeonseo Kim, Kateryne Ccama Mamani, In S. Kim, and Heechul Choi. "Facile fabrication of robust and tight PIMs-based TFC hollow fiber membranes with a semi-interpenetrating polymer network via liquid phase cross-linking for organic solvent nanofiltration." *Journal of Membrane Science* 713 (2025): 123338.

Yejin Liang, **Minbeom Kim**, Eunmok Yang, Jaehyun Byun, Hyeonseo Kim, Kateryne Ccama Mamani, In S. Kim, and Heechul Choi. "Tailoring asymmetric wettability of Janus nanofiber membranes for directional oil transport and oil capture from oil/water mixtures and oil-in-water emulsions." *Chemical Engineering Journal* 496 (2024): 154087.

Eunmok Yang, **Minbeom Kim**, Yejin Liang, Jaehyun Byun, Hyeonseo Kim, Jun Kim, and Heechul Choi. Tailoring "Tailoring the pore size of intermolecular cross-linked PIMs-based thin-film composite hollow fiber membranes using different length cross-linkers for organic solvent nanofiltration." *Chemical Engineering Journal* 474 (2023): 145339.

Yejin Liang, Eunmok Yang, **Minbeom Kim**, Soyoung Kim, Hyeonseo Kim, Jaehyun Byun, Numan Yanar, and Heechul Choi. "Lotus leaf-like SiO2 nanofiber coating on polyvinylidene fluoride nanofiber membrane for water-in-oil emulsion separation and antifouling enhancement." *Chemical Engineering Journal* 452 (2023): 139710.

Numan Yanar, Yejin Liang, Eunmok Yang, **Minbeom Kim**, Hyeonseo Kim, Jaehyun Byun, Moon Son, and Heechul Choi. "Robust and fouling-resistant ultrathin membranes for water purification tailored via semi-dissolved electrospun nanofibers." *Journal of Cleaner Production* 418 (2023): 138056.

Eunmok Yang, Yejin Liang, Numan Yanar, **Minbeom Kim**, Hosik Park, and Heechul Choi. "Intermolecular

cross-linked polymer of intrinsic microporosity-1 (PIM-1)-based thin-film composite hollow fiber membrane for organic solvent nanofiltration." *Journal of Membrane Science* 671 (2023): 121370.

Sung Hee Joo, Yejin Liang, **Minbeom Kim**, Jaehyun Byun, and Heechul Choi. "Microplastics with adsorbed contaminants: Mechanisms and Treatment." *Environmental Challenges* 3 (2021): 100042.

Patent

Heechul Choi and **Minbeom Kim**. "Necklace-like nanofibrous hybrid membrane for simultaneous removal of particulate matter and sulfur dioxide and method for preparing the same" *Korean Patent (registered, 2024) and U.S. patent (registered, 2026)*

Technology Transfer

Heechul Choi and **Minbeom Kim**. "Necklace-like nanofibrous hybrid membrane for simultaneous removal of particulate matter and sulfur dioxide and method for preparing the same" (2024), *USD 72,000*

HONERS AND AWARDS

Full Government Scholarship (Ph.D. Program), GIST	Mar 2020 – Aug 2025
RA (former GUP) Scholarship, GIST	Mar 2021 – Feb 2022
Full Government Scholarship (M.S. Program), GIST	Mar 2018 – Feb 2020
Full Government Scholarship (B.S. Program), GIST	Mar 2014 – Feb 2018
Best Presentation Award, Korean Society of Environmental Engineers	Nov 2025
Best Presentation Award, Korean Society of Environmental Engineers	Nov 2026

SKILLS AND INTERESTS

Instrumentation & measurement techniques

Completed certified user training and granted direct (Super User) access to operate the following instruments independently: Scanning electron microscopy (SEM), Atomic force microscope (AFM), X-ray diffraction (XRD), Brunauer-Emmett-Teller (BET), Fourier transform infrared spectroscopy (FT-IR), X-ray photoelectron spectroscopy (XPS), UV-Vis spectrophotometer, Liquid chromatography-mass spectrometry/mass spectrometry (LC-MS/MS), Inductively coupled plasma mass spectrometry (ICP-MS), Universal testing machine (UTM), Thermal gravimetric analyzer (TGA), Zeta potential analyzer (DLS), Raman spectroscopy, Four point probe (sheet resistivity analyzer), and Water contact angle (WCA) analyzer, Dynamic vapor sorption (DVS) analyzer, Electrochemical characterization (potentiostat-based systems).

Experimental skills

- Fabrication of nanofiber (electro -spinning, -spraying), hollow fiber, flat-sheet, TFC membranes.
- Functionalization of membrane by porous nanomaterials (MOFs, carbon-based, porous polymer).
- Installing and operating lab-scale pressure- and gravity-driven membrane systems for environmental pollution control and liquid/gas/ion separation (with ongoing electrochemical characterization training).

Languages

English (Fluent), Korean (Native), Mandarin Chinese (Basic)

Interests

Team Sports (especially Soccer, Badminton, and Golf), Traveling